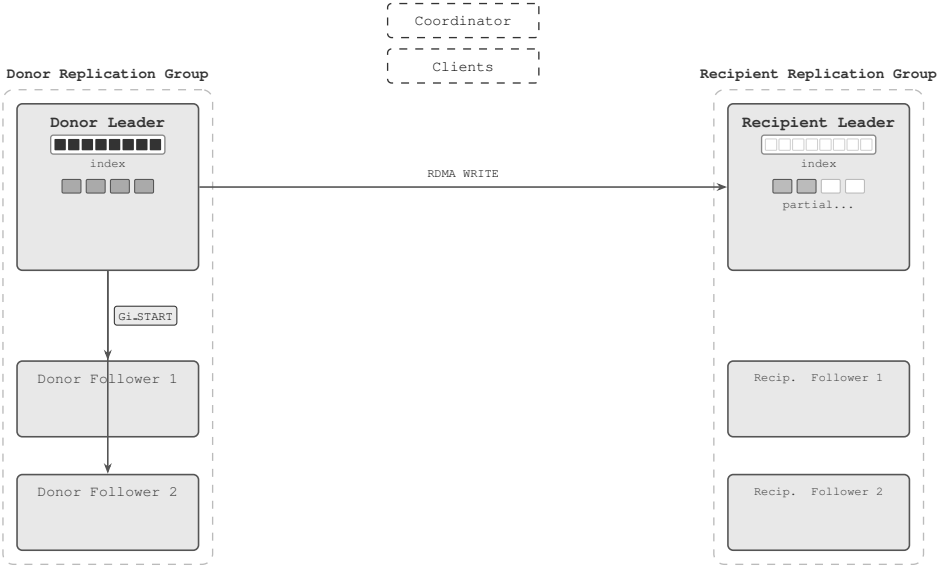
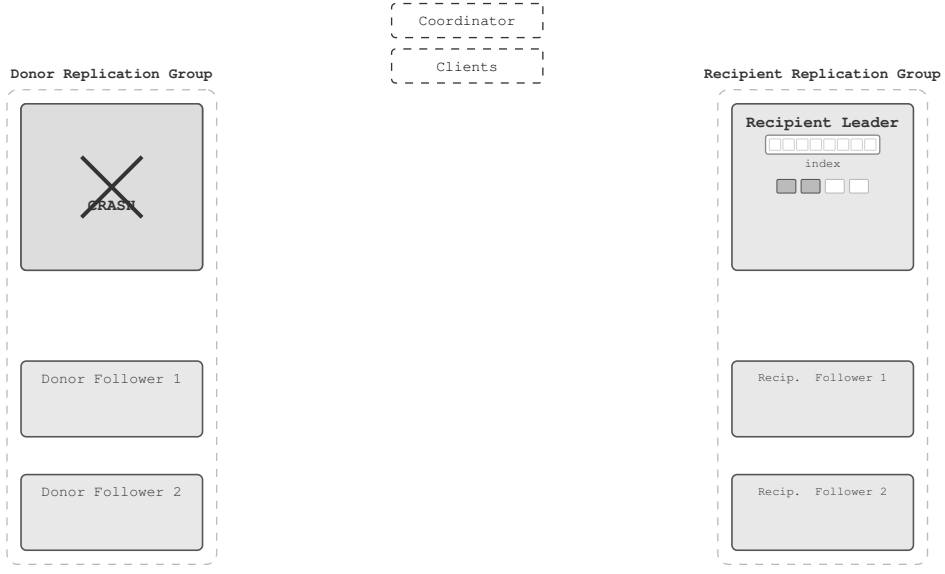
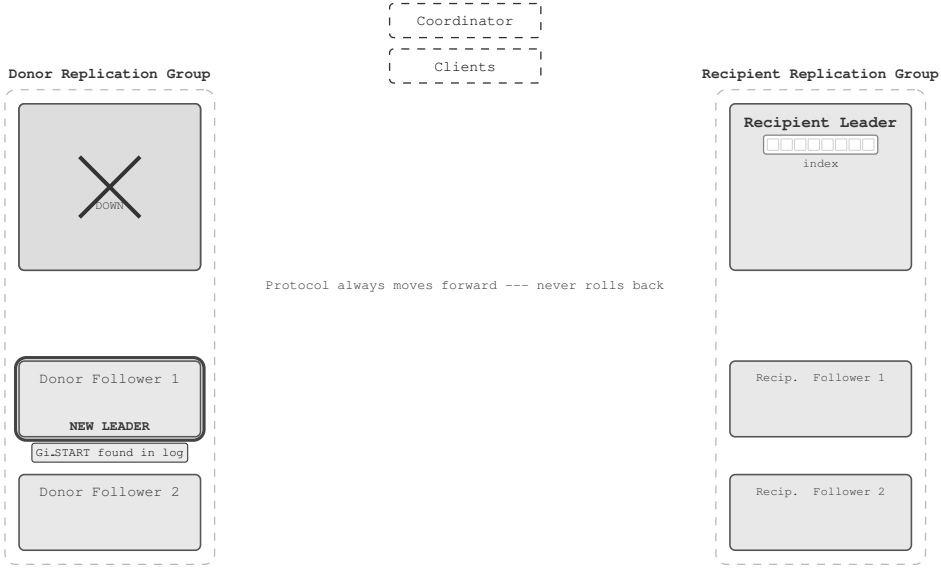
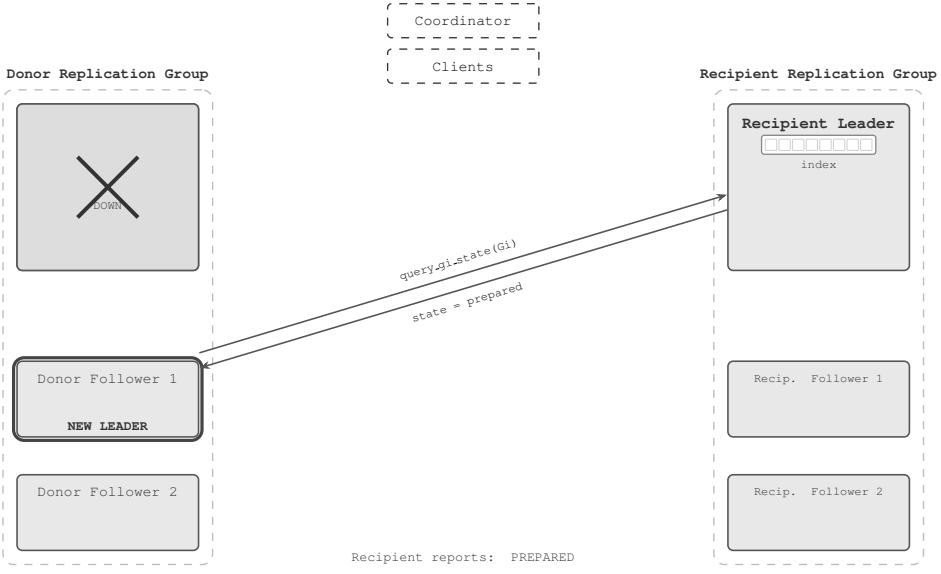
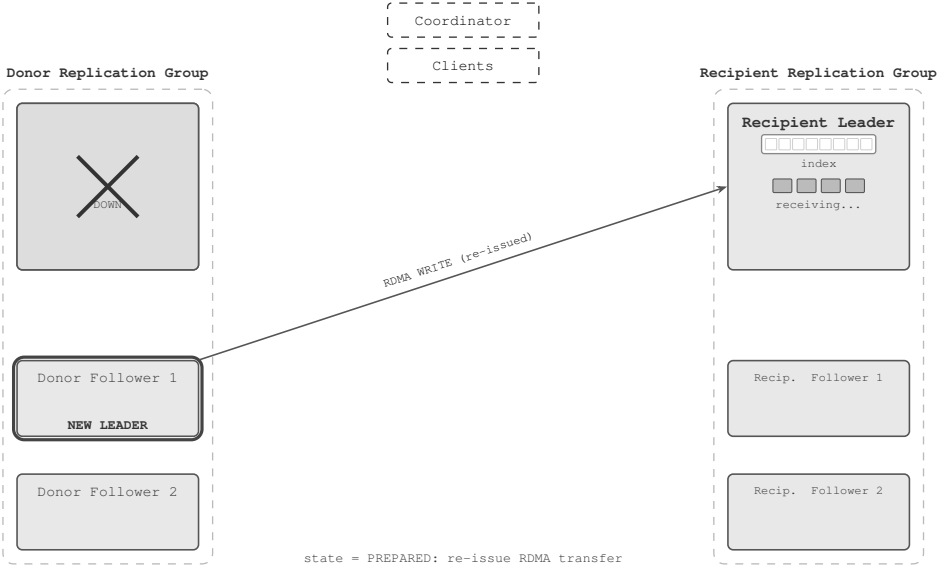
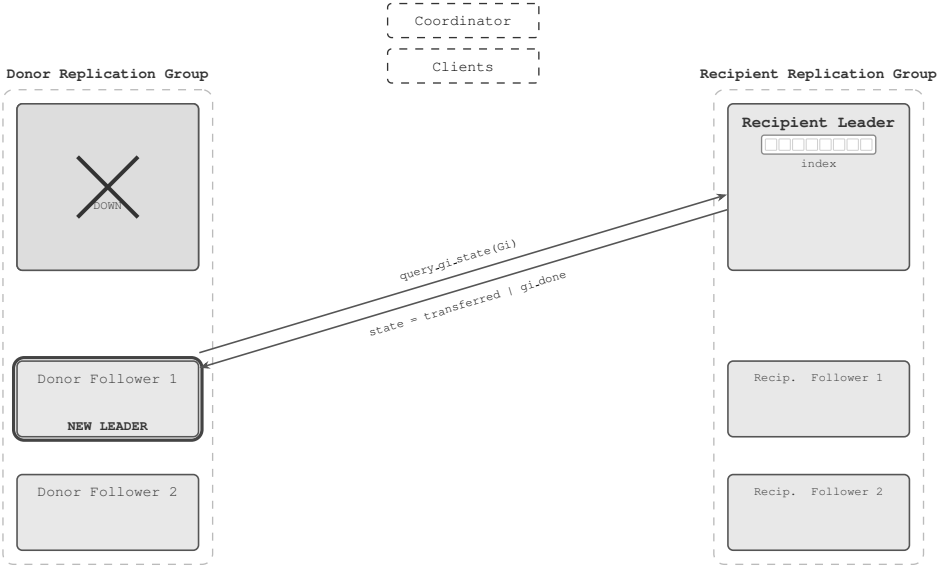

Coordinator
Clients
Donor Replication Group
Donor Leader
index
RDMA WRITE
Recipient Replication Group
Recipient Leader
index
partial...
Gi_START
Donor Follower 1
Donor Follower 2
Recip. Follower 1
Recip. Follower 2

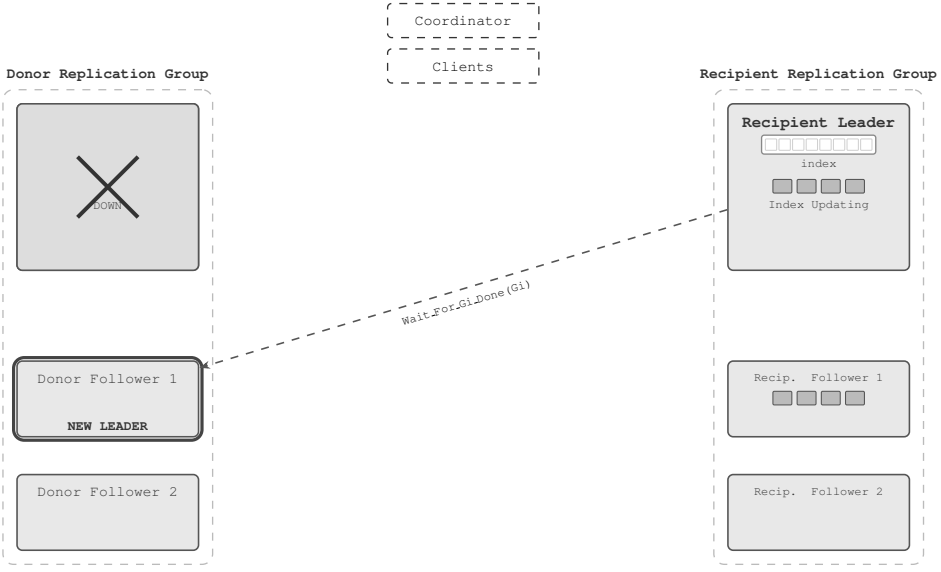

Coordinator
Clients
Donor Replication Group
CRASH
Donor Follower 1
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2

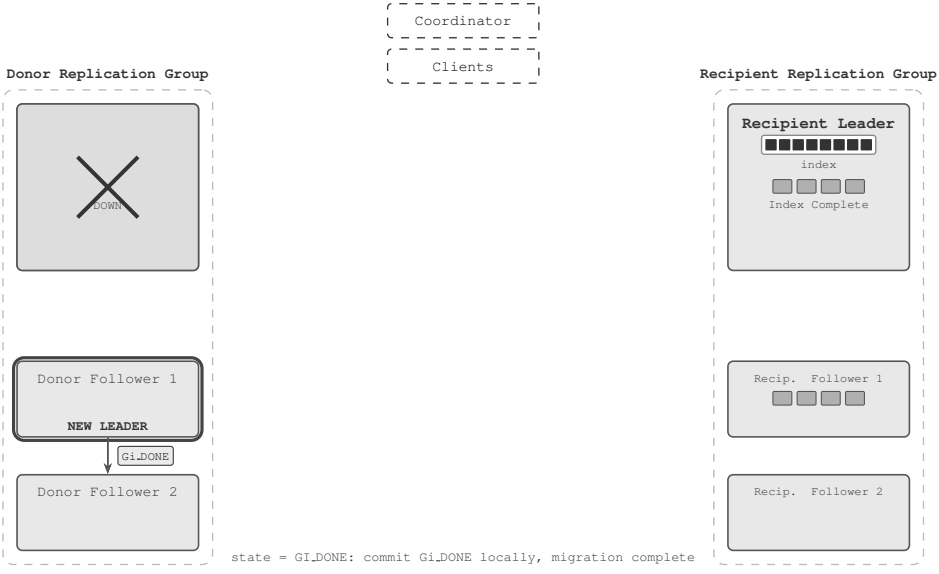

Coordinator
Clients
Donor Replication Group
DOWN
Donor Follower 1
NEW LEADER
Gi_START found in log
Donor Follower 2
Protocol always moves forward --- never rolls back
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2


Coordinator
Clients
Donor Replication Group
Recipient Replication Group
DOWN
Recipient Leader
index
query_gi_state(Gi)
state = prepared
Donor Follower 1
NEW LEADER
Donor Follower 2
Recip. Follower 1
Recip. Follower 2
Recipient reports: PREPARED


Coordinator
Clients
Donor Replication Group
DOWN
Donor Follower 1
NEW LEADER
Donor Follower 2
RDMA WRITE (re-issued)
Recipient Replication Group
Recipient Leader
index
receiving...
Recip. Follower 1
Recip. Follower 2
state = PREPARED: re-issue RDMA transfer


Coordinator
Clients
Donor Replication Group
DOWN
Donor Follower 1
NEW LEADER
Donor Follower 2
query_gi_state(Gi)
state = transferred | gi_done
Recipient Replication Group
Recipient Leader
index
Recip. Follower 1
Recip. Follower 2


Coordinator
Clients
Donor Replication Group
DOWN
Donor Follower 1
NEW LEADER
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Index Updating
Wait For Gi Done(Gi)
Recip. Follower 1
Recip. Follower 2


Coordinator
Clients
Donor Replication Group
DOWN
Donor Follower 1
NEW LEADER
Gi_DONE
Donor Follower 2
Recipient Replication Group
Recipient Leader
index
Index Complete
Recip. Follower 1
Recip. Follower 2
state = GI_DONE: commit Gi_DONE locally, migration complete